

HTML & CSS Foundations Quiz

Test your fundamental knowledge of webpage structure, semantic elements, layout models, and CSS styling.

Name: _____

Date: _____

Time Allowed: 20 Minutes

Total Questions: 10

Q1. Which HTML5 element should be used to encapsulate self-contained, reusable content like blog posts or news items?

- ☐ A) `<section>`
- ☐ B) `<article>`
- ☐ C) `<aside>`
- ☐ D) `<div>`

Q2. According to the standard CSS Box Model, what components make up an element's total rendered width?

- ☐ A) Content + Padding + Border
- ☐ B) Content + Margin
- ☐ C) Content + Padding + Border + Margin
- ☐ D) Padding + Border + Outline

Q3. Which CSS property is used to change the box model calculation so that `width` includes padding and border?

- ☐ A) `box-sizing: content-box;`
- ☐ B) `box-sizing: border-box;`
- ☐ C) `box-layout: fixed;`
- ☐ D) `display: inline-block;`

Q4. Which selector has the highest CSS specificity?

- ☐ A) `#header .nav-item` (1 ID, 1 Class)
- ☐ B) `div.header ul li.active` (2 Classes, 3 Elements)
- ☐ C) `#header #main-nav` (2 IDs)
- ☐ D) `header nav a:hover` (1 Pseudo-class, 3 Elements)

Q5. How does an element with `position: absolute;` position itself?

- ☐ A) Relative to its normal flow position
- ☐ B) Relative to the browser viewport always
- ☐ C) Relative to its nearest positioned ancestor
- ☐ D) Relative to the root `<html>` tag only

Q6. Which CSS Flexbox property is used to align flex items along the main axis?

- ☐ A) `align-items`
- ☐ B) `justify-content`
- ☐ C) `align-content`
- ☐ D) `flex-direction`

Q7. What is the correct HTML attribute to specify alternative text for an image element?

- ☐ A) `title="description"` ☐ B) `alt="description"`
☐ C) `src="description"` ☐ D) `caption="description"`

Q8. In CSS Grid, which shorthand property creates 3 equal-width columns?

- ☐ A) `grid-template-columns: repeat(3, 1fr);` ☐ B) `grid-columns: 3 auto;`
☐ C) `flex-grid: 1fr 1fr 1fr;` ☐ D) `grid-template-rows: repeat(3, 33%);`

Q9. Which unit is relative to the font-size of the root (`<html>`) element?

- ☐ A) `em` ☐ B) `rem`
☐ C) `vh` ☐ D) `px`

Q10. Which meta tag ensures proper responsive scaling on mobile browser viewports?

- ☐ A) `<meta name="responsive" content="true">` ☐ B) `<meta name="viewport" content="width=device-width, initial-scale=1.0">`
☐ C) `<meta screen="mobile" scale="1.0">` ☐ D) `<meta display="mobile-first">`

Answer Key & Explanations

Q1: B (<article>)

The `<article>` tag represents a self-contained composition intended to be independently reusable (e.g., forum posts, magazine articles). `<section>` represents a general standalone section of content.

Q2: C (Content + Padding + Border + Margin)

The CSS Box Model consists of four concentric boxes surrounding content: Content area, Padding, Border, and outer Margin.

Q3: B (box-sizing: border-box;)

Setting `box-sizing: border-box` forces the element's explicit `width` and `height` to include content, padding, and border, preventing layout overflow issues.

Q4: C (#header #main-nav)

Specificity calculation uses weight tiers: (IDs, Classes/Attributes, Elements). `#header #main-nav` has a score of (2, 0, 0), making it more specific than option A (1, 1, 0) or option B (0, 2, 3).

Q5: C (Relative to its nearest positioned ancestor)

An absolute element is removed from normal flow and positioned relative to its closest parent with a position value other than `static` (or the initial containing block if none exists).

Q6: B (justify-content)

`justify-content` controls item alignment along the main axis (horizontal by default), whereas `align-items` controls alignment along the cross axis.

Q7: B (alt="description")

The `alt` attribute provides descriptive text for screen readers or when the image fails to load. The `title` attribute only provides a visual tooltip hover.

Q8: A (grid-template-columns: repeat(3, 1fr);)

The `repeat(3, 1fr)` expression expands to `1fr 1fr 1fr`, dividing available horizontal grid container space into 3 equal flexible fraction tracks.

Q9: B (rem)

`rem` stands for "root em" and scales relative strictly to the font size set on the root `<html>` element. The standard `em` scales relative to its parent element's font size.

Q10: B (<meta name="viewport" content="width=device-width, initial-scale=1.0">)

This viewport meta tag instructs mobile browsers to render the document width to equal the physical screen size, disabling automatic mobile desktop-rendering zoom.